

Patient brief — Yujie Cao (DOB 8 Aug 1997, 29F)

For: Dr David Wilson, neurosurgery consult, RPA — Thursday 27 Aug 2026, 12:30pm. Prepared by the patient and her partner to supplement the referral. Imaging: I-MED Chatswood, Acc #77.60502003.

Diagnosis (MRI lumbar spine + sacrum, 18 Aug 2026, Dr U Ridley)

Conus at **L4**, tethered to a **121 × 49 × 20 mm cauda equina lipoma** (T1/T2-hyperintense, fat-suppressing, non-enhancing); deformed posterior conus; lipoma extends into the **left neural foramen encasing the left L5 root**; left lateral-recess effacement **compressing the left S1 ganglion; enhancing 16 × 16 mm subcutaneous mass at S1 suggestive of a dermal sinus**; no diastematomyelia; no disc protrusion or stenosis.

What the referral letter doesn't include

1. **Lifelong (congenital-pattern) sphincter picture** — severe constipation since early childhood (bowel motions every 4–6 days; childhood retention requiring parental intervention), lifelong urinary frequency/urge incontinence, anorgasmia with reduced genital sensation. Not new — lifelong.
2. **Lifelong left-leg asymmetry** — left leg always colder; left calf/thigh thinner; leg slightly shorter; **unilateral LEFT pes cavus with claw toes** (photos on phone); left toe-walk/heel-walk weaker. Circumferences measured at home: calf L ____ / R ____ cm; thigh L ____ / R ____ cm.
3. **Sacral cutaneous marker** — skin mark over the sacrum (photo on phone), corresponding to the dermal sinus on MRI. No discharge/redness to date; we are not touching it.
4. **12-month progressive pain, single continuous LEFT strip** — from the left vaginal margin (S3) → posterior thigh (S2) → heel (S1). Spontaneous electric shocks proximally, relieved by pressure, worst lying down. Onset ~Aug 2025, now more constant and **worsening**, with new toe/heel-walk weakness.
5. **Recent bowel-pattern change (Aug 2026)** — from lifelong constipation to episodic loose stool, possibly with reduced awareness — coincident with the worsening leg symptoms.

Urodynamics (8 Oct 2025, Sydney Urodynamic Centres)

Peak flow 12 ml/s, intermittent; **residual 60 → 300 ml** by end of study; first sensation 100 ml; **capacity 300 ml; several episodes of detrusor overactivity with leakage**; urethral closure normal (63 cmH2O). Read with the MRI: neurogenic pattern. Betmiga 50 mg was prescribed but **never taken**. She takes **no regular medications** (never OCP/SSRI). Nulliparous.

Questions

1. Is detethering surgery indicated, and how urgent given the documented progression (pain, new weakness, bowel change)?
2. What operation would you plan for this anatomy — radical lipoma resection vs partial debulk + release? How many adult cauda-equina-lipoma detethers do you/your unit do?
3. Would intraoperative neuromonitoring be used?
4. The dermal sinus — would the tract be excised at the same operation? Any precautions in the meantime beyond not touching it and the fever/discharge red flags?
5. Likely waiting-list category and realistic surgery date at RPA?
6. Is a whole-spine MRI needed (second tether/syrinx)?
7. Main risks for her: CSF leak, neurological worsening, re-tether rate?
8. Bladder prognosis — is the neurogenic pattern likely to stabilise or improve? Should neuro-urology be involved now (300 ml residuals, drug choices)?
9. She hopes to have children — should surgery precede conception, and what interval would you advise? (We understand 5 mg folate is recommended.)
10. Anything she should do or avoid while waiting?